

Besant Technologies

PROJECT REPORT

PYTHON

MINI GAMES HUB

A Collection of Fun Games Built with Python



Rock Paper
Scissors



Number
Guessing Game



Dice Rolling
Game



Coin Toss
Game



Mini
Cricket



Penalty
Shootout



Python
Quiz



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GITHUB REPOSITORY
<https://github.com/Priyanka-hati/Python-Mini-Games-Hub>

Small Games... Big Learning!

2. ACKNOWLEDGEMENT

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3. INDEX

Section	Page No.
1. Cover Page	1
2. Acknowledgement	2
3. Index	3
4. Introduction to Python	4
5. Introduction to Project	5
6. Concepts Used in Project	6
7. Source Code	8
8. Working Flow of Code	11
9. Output	13
10. Conclusion	18
11. Bibliography	19

4. INTRODUCTION TO PYTHON

Python is a high-level, general-purpose programming language known for its simple syntax, readability, and wide range of applications. It was created by Guido van Rossum and first released in 1991. Python supports procedural, object-oriented, and functional programming styles.

One of Python's major strengths is its extensive standard library and large ecosystem of third-party packages. Python is widely used in web development, automation, data analysis, artificial intelligence, machine learning, scientific computing, education, and application development. Its readable syntax makes it suitable for beginners while also supporting complex software projects.

Python is an interpreted language and provides built-in data structures such as lists, tuples, sets, and dictionaries, along with control statements, functions, classes, and exception handling.

Python is useful for learning programming because concepts can be demonstrated with relatively small programs. In this project, Python is used to create a menu-driven collection of mini games and demonstrate input/output, conditions, loops, functions, classes and objects, random number generation, collections, validation, and score calculation.

5. INTRODUCTION ABOUT PROJECT

Python Mini Games Hub is a console-based, menu-driven application developed using Python 3. The project combines seven small interactive games into a single application. The user selects a game from the main menu, plays the selected game, receives a result or score, and can return to the main menu.

The project uses Python's random module to introduce unpredictable computer-generated values. `random.choice()`, `random.randint()`, and `random.shuffle()` are used in different games. This makes the application interactive while providing practical examples of randomization.

The seven games are Rock Paper Scissors, Number Guessing Game, Dice Rolling Game, Coin Toss Game, Mini Cricket Game, Penalty Shootout, and Python Quiz. Each game focuses on different programming concepts such as conditional logic, loops, validation, collections, randomization, score calculation, and exception handling.

The main objective is educational: to demonstrate how basic and intermediate Python concepts can be combined to create a complete application. The project also provides practice in object-oriented programming through the `MiniGamesHub` class.

6. CONCEPTS USED IN PROJECT

6.1 Variables and Data Types

Variables store values such as scores, choices, attempts, and random values. The project uses integers, strings, lists, and dictionaries.

6.2 Input and Output

`input()` accepts user choices, while `print()` displays menus, instructions, scores, and results.

6.3 Conditional Statements

`if`, `elif`, and `else` compare choices, validate values, and determine results.

6.4 Loops

`while` loops support repeated interaction and validation. `for` loops handle penalty rounds and quiz questions.

6.5 Functions and Methods

Each game is implemented as a separate method inside `MiniGamesHub`, improving modularity.

6.6 Lists

Lists store game choices such as Rock, Paper, Scissors; Heads, Tails; and penalty directions.

6.7 Dictionaries

Quiz questions are stored as dictionaries containing a question, options, and answer.

6.8 Strings

Strings are used for menus, messages, choices, and results. `strip()` and `upper()` normalize input.

6.9 Random Module

`random.choice()` selects an item, `random.randint()` generates an integer, and `random.shuffle()` changes question order.

6.10 Exception Handling

`try` and `except ValueError` prevent the program from stopping when invalid numeric input is entered.

6.11 Object-Oriented Programming

The `MiniGamesHub` class groups game methods and score attributes into one object-oriented structure.

6.12 Menu-Driven Programming

`main_menu()` repeatedly displays options and calls the selected game until Exit is chosen.

7. SOURCE CODE

The application is implemented in Jupyter Notebook using three logical cells: importing the required module, defining the MiniGamesHub class, and creating an object to start the main menu. The actual full source code should be inserted as screenshots from the executed notebook.

7.1 Required Module

```
import random
```

7.2 Class and Game Methods

```
: import random

: class MiniGamesHub:

    def __init__(self):
        self.player_score = 0
        self.computer_score = 0

    # -----
    # 1. Rock Paper Scissors
    # -----
    def rock_paper_scissors(self):

        print("\n===== ROCK PAPER SCISSORS =====")

        choices = ["rock", "paper", "scissors"]

        player = input("Enter rock, paper, or scissors: ").lower()

        if player not in choices:
            print("Invalid choice!")
            return

        computer = random.choice(choices)

        print("Computer chose:", computer)

        if player == computer:
            print("It's a tie!")

        elif (
            (player == "rock" and computer == "scissors")
            or (player == "paper" and computer == "rock")
            or (player == "scissors" and computer == "paper")
        ):
            print("You win!")
            self.player_score += 1

        else:
            print("Computer wins!")
            self.computer_score += 1

        print("Player Score:", self.player_score)
        print("Computer Score:", self.computer_score)
```

```

# -----
# 2. Number Guessing Game
# -----
def number_guessing(self):

    print("\n===== NUMBER GUESSING GAME =====")

    number = random.randint(1, 10)

    attempts = 3

    for attempt in range(1, attempts + 1):

        try:
            guess = int(input("Guess a number between 1 and 10: "))

            if guess == number:
                print("Congratulations! You guessed correctly!")
                self.player_score += 1
                break

            elif guess < number:
                print("Too low!")

            else:
                print("Too high!")

            print("Attempts remaining:", attempts - attempt)

        except ValueError:
            print("Please enter a valid number.")

    else:
        print("Game Over!")
        print("The correct number was:", number)

```

```

# -----
# 3. Dice Rolling Game
# -----
def dice_rolling(self):

    print("\n===== DICE ROLLING GAME =====")

    input("Press Enter to roll the dice...")

    player_dice = random.randint(1, 6)
    computer_dice = random.randint(1, 6)

    print("You rolled:", player_dice)
    print("Computer rolled:", computer_dice)

    if player_dice > computer_dice:
        print("You win!")
        self.player_score += 1

    elif player_dice < computer_dice:
        print("Computer wins!")
        self.computer_score += 1

    else:
        print("It's a tie!")

# -----
# 4. Coin Toss Game
# -----
def coin_toss(self):

    print("\n===== COIN TOSS GAME =====")

    choice = input("Choose Heads or Tails: ").lower()

    if choice not in ["heads", "tails"]:
        print("Invalid choice!")
        return

    result = random.choice(["heads", "tails"])

    print("Coin result:", result)

    if choice == result:
        print("You win!")
        self.player_score += 1

    else:
        print("Computer wins!")
        self.computer_score += 1

```

```

# -----
# 5. Mini Cricket Game
# -----
def mini_cricket(self):

    print("\n===== MINI CRICKET GAME =====")

    print("You have 3 balls.")
    print("Choose runs between 1 and 6.")

    player_runs = 0

    for ball in range(1, 4):

        try:
            player = int(input(f"Ball {ball} - Enter your runs (1-6): "))

            if player < 1 or player > 6:
                print("Please enter a number between 1 and 6.")
                continue

            computer = random.randint(1, 6)

            print("Computer bowled:", computer)

            if player == computer:
                print("OUT!")
                break

            player_runs += player
            print("Runs scored:", player)

        except ValueError:
            print("Invalid input!")

    print("Your total score:", player_runs)

    computer_runs = random.randint(1, 15)

    print("Computer score:", computer_runs)

    if player_runs > computer_runs:
        print("You win!")
        self.player_score += 1

    elif player_runs < computer_runs:
        print("Computer wins!")
        self.computer_score += 1

    else:
        print("Match tied!")

```

```

# -----
# 6. Penalty Shootout
# -----
def penalty_shootout(self):

    print("\n==== PENALTY SHOOTOUT =====")

    player_score = 0
    computer_score = 0

    for round_no in range(1, 4):

        print("\nRound", round_no)

        direction = input(
            "Choose left, center, or right: "
        ).lower()

        if direction not in ["left", "center", "right"]:
            print("Invalid direction!")
            continue

        goalkeeper = random.choice(
            ["left", "center", "right"]
        )

        print("Goalkeeper chose:", goalkeeper)

        if direction == goalkeeper:
            print("Saved!")

        else:
            print("GOAL!")
            player_score += 1

    computer_score = random.randint(0, 3)

    print("\nYour goals:", player_score)
    print("Computer goals:", computer_score)

    if player_score > computer_score:
        print("You win!")
        self.player_score += 1

    elif player_score < computer_score:
        print("Computer wins!")
        self.computer_score += 1

    else:
        print("It's a draw!")

```

```

# -----
# 7. Python Quiz
# -----
def python_quiz(self):

    print("\n----- PYTHON QUIZ -----")

    questions = [
        {
            "question": "Which keyword is used to define a function?",
            "options": ["A. function", "B. def", "C. fun", "D. define"],
            "answer": "b"
        },
        {
            "question": "Which symbol is used for comments in Python?",
            "options": ["A. //", "B. /*", "C. #", "D. --"],
            "answer": "c"
        },
        {
            "question": "Which function is used to display output?",
            "options": ["A. display()", "B. print()", "C. show()", "D. output()"],
            "answer": "b"
        },
        {
            "question": "Which data type stores True or False?",
            "options": ["A. int", "B. string", "C. bool", "D. float"],
            "answer": "c"
        },
        {
            "question": "Which function is used to get input from the user?",
            "options": ["A. input()", "B. get()", "C. read()", "D. scan()"],
            "answer": "a"
        }
    ]

    score = 0

    for question in questions:

        print("\n" + question["question"])

        for option in question["options"]:
            print(option)

        answer = input("Enter your answer: ").lower()

        if answer == question["answer"]:
            print("Correct!")
            score += 1

        else:
            print("Wrong!")

    print("\nQuiz completed!")
    print("Your score:", score, "/", len(questions))

    if score >= 3:
        print("Good job!")
        self.player_score += 1
    else:
        print("Keep practicing!")

```

```

# -----
# Main Menu
# -----
def main_menu(self):

    while True:

        print("\n-----")
        print("      MINI GAMES HUB")
        print("-----")

        print("1. Rock Paper Scissors")
        print("2. Number Guessing Game")
        print("3. Dice Rolling Game")
        print("4. Coin Toss Game")
        print("5. Mini Cricket Game")
        print("6. Penalty Shootout")
        print("7. Python Quiz")
        print("8. Show Score")
        print("9. Exit")

        print("-----")

        choice = input("Enter your choice (1-9): ")

        if choice == "1":
            self.rock_paper_scissors()

        elif choice == "2":
            self.number_guessing()

        elif choice == "3":
            self.dice_rolling()

        elif choice == "4":
            self.coin_toss()

        elif choice == "5":
            self.mini_cricket()

        elif choice == "6":
            self.penalty_shootout()

        elif choice == "7":
            self.python_quiz()

        elif choice == "8":
            print("\n----- SCORE -----")
            print("Player Score:", self.player_score)
            print("Computer Score:", self.computer_score)

        elif choice == "9":
            print("\nThank you for playing Mini Games Hub!")
            print("Exiting application...")
            break

        else:
            print("Invalid choice! Please select 1-9.")

game_hub = MiniGamesHub()
game_hub.main_menu()

```

7.3 Starting the Application

```

game_hub = MiniGamesHub()
game_hub.main_menu()

```

8. WORKING FLOW OF CODE

The application starts by importing the random module. A MiniGamesHub object is created and main_menu() is called. The menu remains active inside a while loop, allowing the user to play multiple games during the same execution.

8.1 Overall Flow

1. Start the application.
2. Import the random module.
3. Create a MiniGamesHub object.
4. Display the main menu.
5. Read the user's menu choice.
6. Call the corresponding game method.
7. Generate random values where required.
8. Accept and validate input.
9. Apply the selected game's rules.
10. Display the result or score.
11. Return to the main menu.
12. Exit when option 8 is selected.

8.2 Game-wise Working

Rock Paper Scissors

The player selects Rock, Paper, or Scissors. The computer chooses randomly and the choices are compared.

Number Guessing Game

A secret integer from 1 to 100 is generated. The player guesses repeatedly and receives Too Low or Too High hints.

Dice Rolling Game

A random value from 1 to 6 is generated for both sides. The higher value wins.

Coin Toss Game

The player selects Heads or Tails and the program randomly generates the coin result.

Mini Cricket

The player selects a number from 1 to 6. Matching the computer's number causes an OUT; otherwise the selected number is added to the score.

Penalty Shootout

Five penalty attempts are played. Matching directions are saves; different directions are goals.

Python Quiz

Quiz dictionaries are shuffled and presented one by one. Correct answers increase the score.

9. OUTPUTS

9.1 Main Menu

```
=====
      MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
```

Enter your choice (1-9):

9.2 Rock Paper Scissors

```
=====
      MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
```

Enter your choice (1-9): 1

```
===== ROCK PAPER SCISSORS =====
```

Enter rock, paper, or scissors: paper

Computer chose: scissors

Computer wins!

Player Score: 0

Computer Score: 1

9.3 Number Guessing Game

```

=====
          MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 2

===== NUMBER GUESSING GAME =====
Guess a number between 1 and 10: 6
Too low!
Attempts remaining: 2
Guess a number between 1 and 10: 9
Too low!
Attempts remaining: 1
Guess a number between 1 and 10: 10
Congratulations! You guessed correctly!

```

9.4 Dice Rolling Game

```

=====
          MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 3

===== DICE ROLLING GAME =====
Press Enter to roll the dice...
You rolled: 2
Computer rolled: 6
Computer wins!

```

9.5 Coin Toss Game

```

=====
          MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 4

===== COIN TOSS GAME =====
Choose Heads or Tails: Tails
Coin result: tails
You win!

```

9.6 Mini Cricket Game

```

=====
          MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 5

===== MINI CRICKET GAME =====
You have 3 balls.
Choose runs between 1 and 6.
Ball 1 - Enter your runs (1-6): 6
Computer bowled: 3
Runs scored: 6
Ball 2 - Enter your runs (1-6): 6
Computer bowled: 3
Runs scored: 6
Ball 3 - Enter your runs (1-6): 6
Computer bowled: 6
OUT!
Your total score: 12
Computer score: 1
You win!

```

9.7 Penalty Shootout

```

=====
          MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 6

===== PENALTY SHOOTOUT =====

Round 1
Choose left, center, or right: center
Goalkeeper chose: right
GOAL!

Round 2
Choose left, center, or right: left
Goalkeeper chose: right
GOAL!

Round 3
Choose left, center, or right: right
Goalkeeper chose: right
Saved!

Your goals: 2
Computer goals: 2
It's a draw!

```

9.8 Python Quiz

```

-----
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 7

===== PYTHON QUIZ =====

Which keyword is used to define a function?
A. function
B. def
C. fun
D. define
Enter your answer: b
Correct!

Which symbol is used for comments in Python?
A. //
B. /*
C. #
D. --
Enter your answer: c
Correct!

Which function is used to display output?
A. display()
B. print()
C. show()
D. output()
Enter your answer: b
Correct!

Which data type stores True or False?
A. int
B. string
C. bool
D. float
Enter your answer: c
Correct!

Which function is used to get input from the user?
A. input()
B. get()
C. read()
D. scan()
Enter your answer: a
Correct!

Quiz completed!
Your score: 5 / 5
Good job!

```

9.9 Invalid Choice

```
=====
      MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 4

===== COIN TOSS GAME =====
Choose Heads or Tails: Head
Invalid choice!
```

```
=====
      MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 10
Invalid choice! Please select 1-9.
```

9.10 Exit

```
=====
      MINI GAMES HUB
=====
1. Rock Paper Scissors
2. Number Guessing Game
3. Dice Rolling Game
4. Coin Toss Game
5. Mini Cricket Game
6. Penalty Shootout
7. Python Quiz
8. Show Score
9. Exit
=====
Enter your choice (1-9): 9

Thank you for playing Mini Games Hub!
Exiting application...
```

10. CONCLUSION

The Python Mini Games Hub project demonstrates how fundamental Python programming concepts can be combined to develop a complete interactive console application. Seven mini games are integrated into a single menu-driven program.

During development, variables, data types, input and output, conditional statements, loops, lists, dictionaries, strings, functions, exception handling, classes, objects, randomization, and score calculation were applied in practical situations.

The project also provides practical experience in object-oriented programming and modular design. Each game is implemented as a separate method, making the application easier to understand and maintain.

Future enhancements could include a graphical user interface, sound effects, difficulty levels, persistent high scores, player profiles, additional games, and a database for storing game history.

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